

SACRAMENTO INTL, CA, USA

WMO: 724839

Lat: 38.696N

Lon: 121.590W

Elev: 7

StdP: 101.24

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 93225

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-0.7	0.9	-6.3	2.2	10.4	-4.0	2.7	10.5	13.2	12.4	11.8	12.3	1.4	320	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	38.0	21.2	36.4	20.6	34.8	19.9	22.6	35.4	21.6	34.1	20.8	32.7	3.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	12.7	28.9	16.9	12.1	27.9	16.0	11.4	26.6	66.2	35.2	62.7	34.3	59.8	32.8	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.7	8.8	7.7	DB	-2.9	41.1	1.9	1.2	-4.3	41.9	-5.4	42.6	-6.5	43.3	-7.9	44.2	
			WB	-3.5	24.2	2.1	1.3	-5.0	25.1	-6.2	25.9	-7.3	26.6	-8.8	27.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	8.6	10.4	12.9	15.2	19.2	22.8	24.9	24.3	22.4	17.7	12.0	8.5
	DBStd	6.58	2.64	2.56	2.89	3.34	3.49	3.30	2.73	2.56	2.83	2.98	3.10	2.99
	HDD10.0	172	58	23	7	2	0	0	0	0	0	0	15	66
	HDD18.3	1375	301	221	169	104	31	3	0	0	3	47	190	306
	CDD10.0	2587	16	36	97	158	284	385	461	444	373	239	75	19
	CDD18.3	749	0	0	1	10	57	138	203	186	126	28	1	0
	CDH23.3	9803	0	0	35	226	844	1822	2532	2267	1586	474	17	0
	CDH26.7	5108	0	0	2	66	367	965	1461	1275	812	160	1	0
Wind	WSAvg	3.4	2.9	3.4	3.5	3.7	3.9	3.9	3.9	3.6	3.1	3.0	2.7	3.1
Precipitation	PrecAvg	448	74	74	62	34	15	4	0	1	3	24	43	99
	PrecMax	619	220	158	188	94	40	36	1	19	14	89	98	240
	PrecMin	158	1	3	0	2	1	0	0	0	0	0	0	0
	PrecStd	126	57	46	51	25	13	9	0	4	4	27	24	71
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.9	21.9	26.5	31.7	36.2	39.5	40.4	39.9	37.6	33.9	25.6	18.4
		MCWB	11.7	14.0	16.7	17.5	20.3	21.7	22.3	21.1	20.1	18.0	15.3	13.1
	2%	DB	16.6	19.3	24.0	28.5	33.3	37.1	37.9	37.4	35.7	30.5	22.9	16.4
		MCWB	11.9	13.6	15.4	16.2	18.7	20.9	21.5	21.0	19.4	16.9	14.6	12.4
	5%	DB	14.9	17.4	22.1	26.1	31.0	34.6	36.2	35.4	33.6	28.1	20.4	15.0
		MCWB	11.4	12.5	14.4	15.6	17.7	19.7	20.9	20.2	18.9	15.9	13.4	11.6
	10%	DB	13.3	15.7	19.7	23.1	28.0	32.3	34.0	33.3	31.8	26.0	17.9	13.7
		MCWB	10.7	11.8	13.4	14.5	16.8	19.1	20.4	19.6	18.3	15.1	12.8	10.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.0	16.0	18.3	19.2	21.6	23.3	24.1	23.4	22.1	19.9	17.3	15.3
		MCDB	16.7	19.2	24.3	28.3	32.4	36.1	37.5	36.8	34.5	30.5	23.4	17.0
	2%	WB	13.3	14.8	16.5	17.6	20.0	21.9	22.8	21.8	20.6	18.2	15.8	13.9
		MCDB	15.1	18.0	22.2	26.0	31.3	34.5	35.6	35.2	32.6	27.7	20.4	15.5
	5%	WB	12.1	13.5	15.2	16.4	18.8	20.8	21.8	20.9	19.7	17.1	14.7	12.7
		MCDB	13.5	16.3	20.4	24.1	29.3	32.5	34.2	33.3	31.5	25.6	18.5	14.0
	10%	WB	11.2	12.4	14.0	15.3	17.5	19.8	20.8	20.2	18.8	16.1	13.7	11.5
		MCDB	12.9	14.8	18.5	22.2	26.9	31.0	32.7	32.0	30.2	23.8	17.1	12.9
Mean Daily Temperature Range	5% DB	MDBR	8.7	9.9	11.8	13.5	15.1	16.8	18.1	18.1	17.4	15.5	11.5	8.9
		MCDBR	12.0	13.2	16.2	18.4	19.4	20.1	20.0	20.4	20.4	19.4	15.5	10.6
		MCWBR	7.6	8.0	8.8	8.4	7.6	7.4	7.6	7.3	7.5	8.2	8.1	6.7
	5% WB	MCDBR	7.8	10.8	14.3	16.5	18.2	18.9	19.2	19.1	18.8	16.8	12.2	6.2
		MCWBR	5.7	7.1	8.3	8.2	7.8	7.7	7.9	7.3	7.3	7.6	7.2	4.6
Clear-Sky Solar Irradiance	taub	0.322	0.325	0.338	0.354	0.364	0.360	0.361	0.344	0.349	0.343	0.327	0.331	
	taud	2.511	2.509	2.461	2.405	2.402	2.420	2.406	2.446	2.442	2.468	2.500	2.479	
	Ebn at Noon	856	904	923	923	915	915	911	921	898	868	841	813	
	Edh at Noon	78	88	102	115	117	115	116	109	103	91	78	75	
All-Sky Solar Radiation	RadAvg	2.16	3.20	4.56	6.06	7.37	8.23	8.20	7.33	5.94	4.18	2.67	1.92	
	RadStd	0.40	0.41	0.48	0.52	0.45	0.33	0.23	0.23	0.21	0.28	0.20	0.32	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (47 neighbors)		N/A	N/A	-1.16	N/A	-0.57	-0.64	N/A	N/A	N/A
		+0.40	N/A	-1.08	N/A	N/A	N/A	-116	+138	+42

Nomenclature: See separate page